

Physik II — FS 2024 — Prof. R. Wallny — Serie 3

Solution

Exercise 1: Boat and satellite

- a) The displacement of the harmonic standing wave which oscillates along the z -axis is given by

$$\xi(x, t) = A_0 \cos(kx) \cos(\omega t), \quad (1)$$

where the wave number is

$$k = \frac{2\pi}{\lambda_{\text{water}}} \quad (2)$$

and angular frequency is

$$\omega = 2\pi\nu = 2\pi \frac{v_{\text{water}}}{\lambda_{\text{water}}}. \quad (3)$$

Since the boat is located at the crest of a wave at $t_0 = 0$, its position as a function of time is:

$$\xi_{\text{crest}}(t) = A_0 \cos(\omega t). \quad (4)$$

- b) The emitter is oscillating harmonically in the vertical direction, i.e. back and forth relative to the satellite. Therefore, we are dealing with the Doppler effect where the observer is at rest and the source is moving: the received frequency is shifted depending on whether the boat is moving towards or away from the satellite. The vertical speed of the boat is given by

$$v_{\text{boat}} = \dot{\xi}_{\text{crest}}(t) = -\omega A_0 \sin(\omega t). \quad (5)$$

Therefore, the maximum speed is $v_{\text{boat}}^{\text{max}} = \omega A_0$, when the boat passes through $\xi_{\text{crest}} = 0$. The maximum frequency will be achieved for the radio wave which is emitted from the boat when the boat is at $\xi_{\text{crest}} = 0$ and moving upwards.

- c) In the reference system of the boat the frequency of the radio wave is $\nu_{\text{radio}} = c/\lambda_{\text{radio}}$, where c is the speed of light. From the Doppler effect, the maximum frequency that the satellite observes is

$$\nu_{\text{satellite}}^{\text{max}} = \frac{\nu_{\text{radio}}}{1 - \frac{v_{\text{boat}}^{\text{max}}}{c}} = \frac{c/\lambda_{\text{radio}}}{1 - \frac{\omega A_0}{c}} = \frac{c/\lambda_{\text{radio}}}{1 - \frac{2\pi v_{\text{water}} A_0}{\lambda_{\text{water}} c}}. \quad (6)$$

Numerically, this yields

$$\nu_{\text{Satellit}}^{\text{max}} \approx (1 + 0.5 \times 10^{-8}) \nu_{\text{radio}} \quad (7)$$

Comparing this to the frequency of the emitted wave

$$\nu_{\text{radio}} = 100 \text{ MHz} \quad (8)$$

shows an increase in frequency of 0.5 Hz.

Remark:

The previous solution is just an approximation that is valid at non-relativistic speeds. The exact solution to this exercise uses the relativistic Doppler effect (equation 5.144 of the lecture notes).

$$\frac{\nu_{\text{Satellit}}^{\text{max}}}{\nu_{\text{radio}}} = \sqrt{\frac{1 + \beta}{1 - \beta}} \quad \text{with} \quad \beta = v_{\text{Schiff}}^{\text{max}}/c_{\text{radio}} \quad (9)$$

With a Taylor expansion of first order around $\beta = 0$, one can show that in the case of small velocities, this equals the classical Doppler effect. With the given values, the relativistic calculation results in a deviation of 1 nHz.

- d) Since the waves are sent isotropically, meaning evenly distributed in all directions of space, the energy flux density on a sphere's surface of given radius around the emitter is constant. At the satellite the radius of such a sphere is z_0 and its surface is $4\pi z_0^2$

The energy flux density at the satellite is

$$S = \frac{P}{4\pi z_0^2}. \quad (10)$$

This is a consequence of energy conservation. See Serie 2, exercise 4.

Exercise 2: Coulomb's law

- a) The superposition of forces created by multiple charges Q_i on a test charge e is given by

$$\vec{F}_{\text{tot}} = \sum_i \vec{F}_{3i} = \frac{e}{4\pi\epsilon_0} \sum_i \frac{Q_i}{|\vec{r}_{3i}|^3} \vec{r}_{3i} \quad (11)$$

The vectors $\vec{r}_{3i}$ between the charge i and the test charge (indexed with a 3) are given by

$$\vec{r}_{31} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, |\vec{r}_{31}| = 5 \quad (12)$$

$$\vec{r}_{32} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, |\vec{r}_{32}| = \sqrt{13} \quad (13)$$

So that the total force on the test charge becomes

$$\vec{F}_{\text{tot}} = \frac{e}{4\pi\epsilon_0} \left(\frac{Q_1}{125} \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \frac{Q_2}{13\sqrt{13}} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) \quad (14)$$

- b) The total charge $Q_{\text{tot}} = Q_1 + Q_2$ is shared uniformly between the two spheres so that both charges become equal to $\frac{Q_{\text{tot}}}{2}$. The force F_{after} after the connection is then given by

$$F_{\text{after}} = \frac{1}{4\pi\epsilon_0} \frac{Q_{\text{tot}}^2}{4r^2} \quad (15)$$

$$\Rightarrow Q_{\text{tot}} = 4r\sqrt{\pi\epsilon_0 F_{\text{after}}} \quad (16)$$

where $r = |\vec{r}_{12}| = 2\mu\text{m}$ denotes the distance between the two spheres. The direction of the force changed after the connection which means the charges had opposite signs before. The force before the connection is then

$$-F_{\text{before}} = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{(Q_{\text{tot}} - Q_{1/2})Q_{1/2}}{r^2} \quad (17)$$

which yields

$$0 = Q_{1/2}^2 - Q_{\text{tot}} Q_{1/2} - 4\pi\epsilon_0 r^2 F_{\text{before}} \quad (18)$$

$$\Rightarrow Q_{1/2} = \frac{Q_{\text{tot}}}{2} \pm \frac{\sqrt{Q_{\text{tot}}^2 + 16\pi\epsilon_0 r^2 F_{\text{before}}}}{2} \quad (19)$$

$$Q_1 = 1.2 \text{ fC} \quad (20)$$

$$Q_2 = -0.4 \text{ fC}. \quad (21)$$

Notice that we assigned a charge to each sphere but that this choice is arbitrary. We know the values of the charges but not which sphere they belong to.

- c) Before the connection, the first sphere held a charge Q_1 and after it, it became a charge $0.5 \cdot Q_{\text{tot}}$. The number of electrons that flowed is then

$$N_{1 \rightarrow 2} = \frac{Q_{1 \rightarrow 2}}{e} = \frac{Q_1 - 0.5 \cdot Q_{\text{tot}}}{e} = 4997 \quad (22)$$

Exercise 3: Spherical layers

- a) From Gauss's law, the following is true

$$\oint_A E_n dA = \frac{q}{\epsilon_0} . \quad (23)$$

where q is the charge contained inside the considered closed volume.

The electrical field within the small layer ($r < r_A$) is parallel to the unit radial vector $\hat{\mathbf{r}}$ because of the spherical symmetry of the problem. The electric field is therefore independent of the solid angle.

$$\mathbf{E}_{r < r_A} = \frac{q_{\text{inside}}}{\epsilon_0 A} \hat{\mathbf{r}} .$$

Additionally, the charge contained within this volume is zero so that the field itself vanishes in that volume: $\mathbf{E}_{r < r_A} = 0$

- b) Gauss's law can be applied also outside both layers.

$$\oint_A E_n dA = \frac{q}{\epsilon_0} , \quad (24)$$

which yields

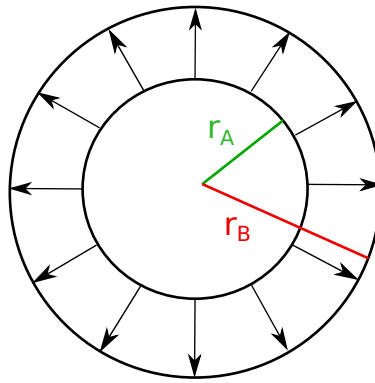
$$\mathbf{E} = \frac{q}{\epsilon_0 A} \hat{\mathbf{r}}$$

because of the spherical symmetry of the problem. The total charge contained in the volume delimited by a radius r such that $r > r_B$ is given by $q = q_A + q_B$. The surface of a sphere at such a radius equals $A = 4\pi r^2$. The field at radius r is then given by

$$\mathbf{E}_{r>r_B} = \frac{q_A + q_B}{\epsilon_0 (4\pi r^2)} \hat{\mathbf{r}} \quad (25)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q_A + q_B}{r^2} \hat{\mathbf{r}} . \quad (26)$$

- c) From the result of exercise b), $\mathbf{E}_{r>r_B} = 0$ is equivalent to having $q_A + q_B = 0$ and $q_A/q_B = -1$.
- d) The following figure shows the field lines for a vanishing field outside of the layers and a positive q_A .

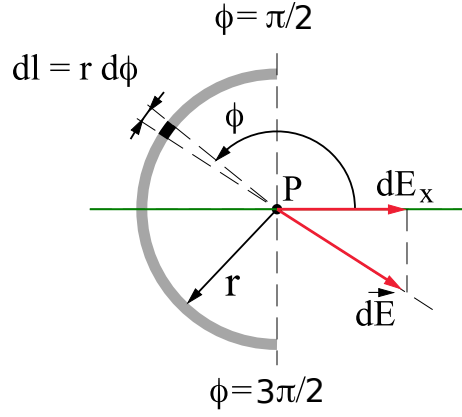


Exercise 4: Semicircular Rod

To solve the problem we are using Coulomb's law for continuous charge distributions ρ . The electric field $\vec{E}$ throughout the entire space is therefore given by

$$\vec{E}(\vec{r}) = \int_V d\vec{E}(V') = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{(\vec{r} - \vec{r}')^2} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|} dV' . \quad (27)$$

Without loss of generality we define the point P as origin of the coordinate system. Hence, $\vec{r} = 0$.



Next, we use the symmetry of the problem to argue that the field is zero in y -direction: The horizontal axis through P (green line in the figure) defines a symmetry axis which means that for every infinitesimally small section of the rod above this axis there is an equally large section below this axis which carries the same charge. Hence, the electric field in every point of the horizontal axis vanishes. Particularly, this means that

$$E_y(\vec{r} = 0) = 0 . \quad (28)$$

Therefore, we will only have to consider the x -component of the electric field in the following:

$$E_x(\vec{r} = 0) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{r}')}{r'^2} \frac{-x'}{|\vec{r}'|} dV' \quad (29)$$

$$= \frac{1}{4\pi\epsilon_0} \int_L \frac{\lambda(\vec{r}')}{r'^2} \frac{-x'}{|\vec{r}'|} dl' , \quad (30)$$

Due to the dimensionality of the problem we here replaced the volume integral over V and the volume charge density ρ by a line integral over the semicircular arch L with line charge density λ . The following applies to the charge dq on the infinitesimally short section dl' of the rod

$$dq = \lambda \cdot dl' = \frac{q}{\pi r} r d\phi . \quad (31)$$

All infinitesimally short sections of the rod are at a distance r from point P . Furthermore,

$$x' = r \cos \phi . \quad (32)$$

Now we integrate in mathematically positive direction (counter-clockwise) along the semicircle from $\pi/2$ to $3\pi/2$. In this way we get

$$E_x(\vec{r} = 0) = \frac{1}{4\pi\epsilon_0} \int_{\pi/2}^{3\pi/2} \frac{q}{r^2 \cdot \pi r} \cdot (-r \cos \phi) d\phi \quad (33)$$

$$= \frac{q}{2\pi^2\epsilon_0 r^2} \quad (34)$$

$$= \frac{10^{-9} \text{ C}}{2\pi^2 \cdot 8.85 \times 10^{-12} \text{ A s V}^{-1} \text{ m}^{-1} \cdot 1 \text{ m}^2} \quad (35)$$

$$= \underline{\underline{5.72 \text{ V m}^{-1}}} . \quad (36)$$

Comment: It is also possible to calculate the electric field in y -direction with Coulomb's law. The calculation is analogous to the calculation we have shown above for the x -component of the electric field. One merely has to replace x' by $y' = r \sin \phi$. One gets

$$E_y(\vec{r} = 0) = \frac{1}{4\pi\epsilon_0} \int_{\pi/2}^{3\pi/2} \frac{q}{r^2 \cdot \pi r} \cdot (-r \sin \phi) d\phi \quad (37)$$

$$= 0 \quad (38)$$

and now also sees mathematically what we have already concluded from the symmetry arguments above: The electric field along the y -direction is zero.

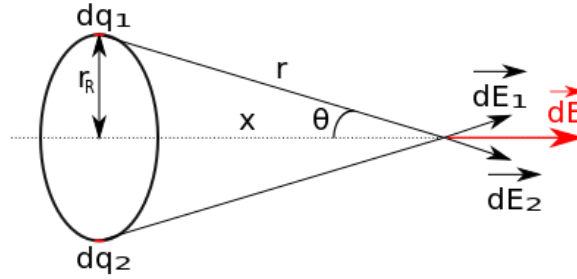
Exercise 5: Oscillation of a charged particle

- a) The electric field E_x on the axis of a ring charge q with radius r_R and at an axial distance x from the ring is given by

$$E_x = \frac{1}{4\pi\epsilon_0} \frac{qx}{(x^2 + r_R^2)^{3/2}}. \quad (39)$$

This results from the following consideration:

We divide the ring in infinitesimally small line elements with charge dq . Due to symmetry, the components of the electric field perpendicular to the symmetry axis of the ring compensate each other as can be seen in the sketch. Because of that, the electric field points along the x -axis.



We consider the line element as a point-like charge and find its contribution to the electric field:

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \cos \theta = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \frac{x}{r} = \frac{1}{4\pi\epsilon_0} \frac{xdq}{(x^2 + r_R^2)^{3/2}} \quad (40)$$

This now has to be integrated over the full charge:

$$E_x = \int dE_x = \int \frac{1}{4\pi\epsilon_0} \frac{xdq}{(x^2 + r_R^2)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{x}{(x^2 + r_R^2)^{3/2}} \int dq = \frac{1}{4\pi\epsilon_0} \frac{qx}{(x^2 + r_R^2)^{3/2}} \quad (41)$$

We transform this by first factoring out r_R in the square root in the denominator:

$$E_x = \frac{1}{4\pi\epsilon_0} \frac{qx}{\left[r_R^2 \left(1 + \frac{x^2}{r_R^2}\right)\right]^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{qx}{r_R^3 \left(1 + \frac{x^2}{r_R^2}\right)^{3/2}} \simeq \frac{1}{4\pi\epsilon_0} \frac{qx}{r_R^3}. \quad (42)$$

The approximation is valid for $x \ll r_R$ (the fraction in the brackets can be neglected).

b) For the force acting on the particle with charge q_0 in x -direction we get

$$F_x = q_0 E_x = \frac{1}{4\pi\epsilon_0} \frac{q_0 q}{r_R^3} x \quad (43)$$

Since the test charge is $q_0 = -q$, this yields

$$F_x = -q E_x = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{r_R^3} x \quad (44)$$

c) After a small displacement of a negatively charged particle, a force arises that is proportional to the displacement and in opposite direction. According to Newton's law,

$$m \frac{d^2 x}{dt^2} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{r_R^3} x \quad (45)$$

This is the differential equation for a harmonic oscillation, which can be generally written as:

$$\frac{d^2 x}{dt^2} = -\omega^2 x \quad (46)$$

d) By comparing the coefficients with equation 46, we can see from the differential equation 45 that the square of the angular frequency of the oscillation is as follows:

$$\omega^2 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{m r_R^3} \quad (47)$$

This results in the following for the angular frequency or the frequency

$$\omega = \sqrt{\frac{1}{4\pi\epsilon_0} \frac{q^2}{m r_R^3}} \quad \text{bzw.} \quad f = \frac{1}{2\pi} \sqrt{\frac{1}{4\pi\epsilon_0} \frac{q^2}{m r_R^3}} \quad (48)$$